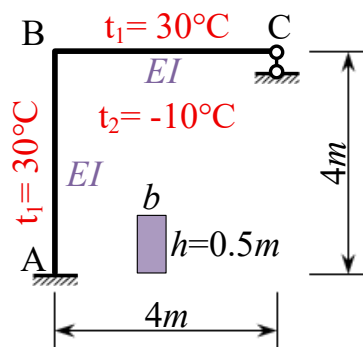




解 (1) 基本位置量

共有 1 个基本未知量:  $\theta_B$

(2) 杆端弯矩

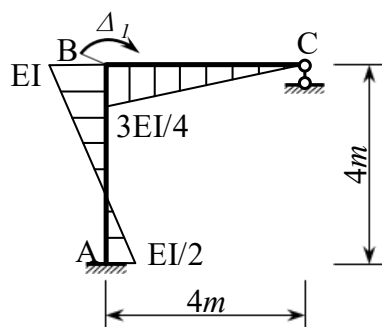
固端弯矩查表求得

$$M_{BA}^F = 4i = \frac{4EI}{l} = EI$$

$$M_{AB}^F = 2i = \frac{2EI}{l} = \frac{EI}{2}$$

$$M_{BC}^F = 3i = \frac{3EI}{l} = \frac{3}{4}EI$$

如图  $\bar{M}$  所示



B 处弯矩平衡得:

$$k_{11} = EI + \frac{3}{4}EI$$

$$k_{11} = \frac{7}{4}EI$$

(3) 温度作用下的杆端弯矩

温度导致的弯曲变形产生的杆端弯矩

$$M_{BA} = M_{AB} = \frac{\alpha EI \Delta t}{h} = 80\alpha EI$$

$$M_{BC} = \frac{3\alpha EI \Delta t}{2h} = 120\alpha EI$$

B 处弯矩平衡得

$$F_{1t}(M) = 40\alpha EI$$

温度导致的纵向的形变量

$$\Delta = \alpha t_0 l = 10\alpha$$

温度导致的轴向变形产生的弯矩

$$M_{BA} = M_{AB} = \Delta \frac{6i}{l} = 15\alpha EI$$

$$M_{BC} = \Delta \frac{3i}{l} = \frac{15}{2}\alpha EI$$

B 处弯矩平衡

$$F_{1t}(N) = \frac{45}{2}\alpha EI$$

求解 $\Delta_1$ :

$$k_{11}\Delta_1 + F_{1t}(M) + F_{1t}(N) = 0$$

解得

$$\Delta_1 = -\frac{250}{7}\alpha$$

根据所求量求解各个点处的弯矩

对柱

$$M_{\text{上}} = -EI * \Delta_1 + 80\alpha EI - 15\alpha EI = \frac{705}{7}\alpha EI$$

$$M_{\text{下}} = \frac{EI}{2} * \Delta_1 + 80\alpha EI + 15\alpha EI = \frac{540}{7}\alpha EI$$

对梁

$$M_B = \frac{3}{4}EI * \Delta_1 + 120\alpha EI + \frac{15}{2}\alpha EI = \frac{705}{7}\alpha EI$$

最终弯矩图如下:

